

GRANBURY, TX, USA

WMO: 720286

Lat: 32.444N

Lon: 97.817W

Elev: 778

StdP: 14.29

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53977

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.2	27.3	10.0	9.4	33.6	14.2	11.6	37.7	24.8	53.2	22.2	50.8	5.1	320	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.5	103.5	72.4	100.4	72.9	98.8	72.7	77.3	90.6	76.4	89.7	75.6	88.8	8.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.9	129.9	82.0	73.1	126.4	81.6	72.3	123.1	81.3	41.1	90.6	40.2	89.3	39.5	89.0	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.9	18.7	16.9	DB	15.9	105.6	4.0	2.7	13.0	107.6	10.6	109.1	8.4	110.6	5.4	112.5
			WB	14.5	78.4	3.9	1.4	11.8	79.4	9.5	80.2	7.3	80.9	4.5	81.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.3	46.8	51.0	59.4	66.6	74.5	82.9	86.0	86.8	79.0	67.9	57.2	48.3	(6)
	DBStd	16.23	9.71	10.98	9.65	8.19	6.98	4.91	4.51	4.76	6.30	8.91	9.58	9.60	(7)
	HDD50	509	174	110	29	4	0	0	0	0	0	3	40	149	(8)
	HDD65	2147	567	404	224	78	13	0	0	0	1	73	265	522	(9)
	CDD50	6821	76	139	321	501	760	987	1115	1142	869	558	256	97	(10)
	CDD65	2984	4	13	51	125	308	537	650	677	419	163	31	6	(11)
	CDH74	37263	60	139	466	1217	3204	6806	8956	9508	4842	1727	274	64	(12)
	CDH80	19685	7	25	113	399	1344	3689	5255	5737	2411	637	59	9	(13)
(14) Wind	WSAvg	7.2	6.9	7.7	8.5	9.0	8.1	7.7	6.7	6.1	5.6	6.4	6.7	6.6	(14)
(15) Precipitation	PrecAvg	35.6	2.2	2.6	3.1	3.1	4.1	3.8	1.9	2.8	3.1	3.6	2.7	2.3	(15)
	PrecMax	54.6	4.2	8.1	7.2	7.0	10.6	11.8	5.2	7.8	8.9	9.5	8.7	10.4	(16)
	PrecMin	19.9	0.6	0.0	0.0	0.4	1.6	1.0	0.0	0.6	0.6	0.4	0.2	0.2	(17)
	PrecStd	8.5	1.0	2.0	1.8	1.7	2.1	3.1	1.4	1.9	2.2	2.4	2.1	2.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.7	83.3	87.4	94.6	97.4	103.5	106.1	106.4	100.4	94.3	86.3	80.9	(19)
		MCWB	57.7	60.9	64.5	67.6	71.9	72.5	71.9	71.9	72.4	70.0	65.7	61.8	(20)
	2%	DB	73.9	78.6	83.3	88.4	93.6	99.7	102.6	103.9	97.1	90.2	81.1	74.5	(21)
		MCWB	57.0	61.3	64.3	66.9	72.0	72.4	72.4	72.6	72.2	70.1	63.9	59.3	(22)
	5%	DB	69.7	73.3	79.4	84.5	90.6	97.3	100.0	101.3	94.7	86.7	76.9	70.1	(23)
		MCWB	56.6	59.9	62.8	66.5	70.9	72.4	72.7	73.3	72.1	68.8	63.1	58.6	(24)
	10%	DB	64.4	69.8	75.3	81.1	87.9	95.0	97.9	99.1	91.6	83.3	72.8	64.8	(25)
		MCWB	53.4	58.7	61.3	65.7	70.4	73.0	72.9	73.0	71.6	66.9	61.5	55.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.0	68.0	69.4	72.7	77.0	78.5	78.2	77.8	77.3	75.3	71.2	67.2	(27)
		MCDB	70.4	77.4	80.4	83.4	89.6	92.2	91.5	91.9	89.8	85.0	79.8	74.6	(28)
	2%	WB	62.2	65.9	67.5	71.3	75.2	76.9	77.1	76.8	75.9	73.5	67.8	64.1	(29)
		MCDB	68.9	73.4	77.5	81.7	86.8	89.9	90.4	91.1	87.4	83.4	74.9	71.1	(30)
	5%	WB	58.7	63.2	66.1	69.9	74.0	75.8	76.2	76.0	75.0	72.2	66.0	60.5	(31)
		MCDB	66.6	70.4	75.4	79.9	85.3	88.7	89.6	90.3	86.2	81.8	73.0	66.5	(32)
	10%	WB	54.3	60.1	64.3	68.4	72.8	75.0	75.3	75.2	74.1	70.6	63.9	56.8	(33)
		MCDB	63.9	67.5	72.7	77.9	83.7	87.9	88.9	89.7	85.4	79.7	71.2	64.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	23.5	22.4	23.0	23.1	20.9	20.5	21.2	21.5	21.3	23.5	23.0	22.4	(35)
		MCDBR	31.3	28.9	26.4	26.9	24.1	23.5	23.6	24.2	23.6	24.7	26.2	29.3	(36)
	5% WB	MCWBR	17.4	16.0	12.2	12.1	8.1	5.3	4.8	4.8	6.0	9.8	13.2	17.1	(37)
		MCWBR	25.8	24.3	21.8	20.4	19.4	19.3	19.8	20.5	19.2	19.6	20.3	22.4	(38)
(40) Clear-Sky Solar Irradiance	taub		0.309	0.319	0.350	0.388	0.421	0.428	0.435	0.428	0.399	0.359	0.334	0.314	(40)
		taud	2.509	2.489	2.420	2.338	2.311	2.325	2.347	2.348	2.374	2.469	2.475	2.498	(41)
	Ebn at Noon		290	298	295	286	276	272	270	271	275	280	278	282	(42)
		Edh at Noon	27	30	35	40	42	41	40	39	36	31	28	26	(43)
(44) All-Sky Solar Radiation	RadAvg		952	1126	1473	1777	1900	2133	2164	1971	1658	1308	1002	838	(44)
	RadStd		97	176	112	148	187	192	144	130	134	196	129	109	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page